FMR

Faithful Medical Reasoning

A trust gate for medical AI answers.

FMR checks whether a medical vision-language model answer is supported by its image, then flags uncertain or uncalibrated outputs for clinician review.

INPUT

Image + question + model answer

CHECK

Image sensitivity + consistency + grounding

ROUTE

Answer or abstain for human review

TEAM: [ADD TEAM NAME BEFORE SUBMISSION]

Purpose: prioritize verification of AI outputs. FMR does not diagnose or estimate patient acuity.

How the review gate works

The system evaluates a model answer; the clinician remains the decision-maker.

1 Medical VLM

Image + question
produces a draft answer

→

2 FMR signals

Image sensitivity
Consistency
Grounding evidence

→

3 Faithfulness

Combine signals;
check calibration

→

4 Review route

ANSWER: show with signals
ABSTAIN: prioritize review

What the checks ask

Would the answer change if image evidence were removed or altered? Is the answer stable across repeated samples? Is image-region evidence available and coherent?

What the output means

A review-priority signal, not a diagnosis, clinical risk score, or proof that an answer is correct. The saved real-data calibration runs currently abstain on all test cases.

Intended setting: clinician review queues for medical AI answers in time-sensitive workflows.

Measured evidence and current limits

LinuxONE execution is verified for the synthetic pipeline; medical-model inference there is not.

LINUXONE | IBM Z / s390x

4,000 synthetic MockVLM cases

5.711 s Python 3.12.15; 2k / 1k / 1k splits

This verifies pipeline execution and artifact generation only. It is not medical inference, clinical validation, or a real-time performance result.

REAL-MODEL PROBES ON LINUXONE

MedVLM-R1: BF16 weights loaded in 102.69 s; a reduced generation call did not finish within 325 s. No answer or FMR score.

SmolVLM-256M (general, not medical): 8 tokens took 119.22 s. Not a practical medical substitute on this CPU VM.

PRIOR GPU / COLAB BENCHMARKS IN THIS REPOSITORY

VQA-RAD (n=114)

MedVLM-R1 47.4% | Qwen2.5-VL-3B 64.0%

PathVQA (n=250)

MedVLM-R1 28.8% | Qwen2.5-VL-3B 27.2%

These baselines were not run on LinuxONE. Saved FMR gate results have 0% test coverage at the chosen error targets.

SLAKE is incomplete: source image assets were missing. Larger calibration data and SLAKE recovery are needed.

Next: finish required labs, replace team placeholder, record a <=3-minute honest demo, and submit the repo + this PDF.